Martin Yuming XU

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Education

MPhil | The Hong Kong Polytechnic University, Dept of Computing 09/2025 - Present
Supervisor: Jason Chen Zhang, Haoyang LI, Ken Man Lung YIU

BSc | South China University of Technology, School of Software 09/2021 - 06/2025
Major: Software Engineering | Minor: Business Administration GPA: 3.66/4.0

Publications

- When Speed meets Accuracy: an Efficient and Effective Graph Model for Temporal Link Prediction
  VLDB 2025
  Haoyang Li, **Yuming Xu**, Yiming Li, Hanmo Liu, Darian Li, Jason Chen Zhang, Lei Chen, Qing Li.
- Towards A Generalizable and Expressive Graph Neural Network for Graph-Level Tasks with Theoretical Guarantees
  VLDB Journal, Jan. 2026. DOI: 10.1007/s00778-026-00978-4
  Luyu Qiu, **Yuming Xu**, Haoyang Li, Jason Chen Zhang, Alexander Zhou, Peng Cheng, Lei Chen, Qing Li.
- Securing Retrieval-Augmented Generation: A Taxonomy of Attacks, Defenses, and Future Directions
  EMNLP 2026
  Yuming Xu, Mingtao Zhang, Zhuohan Ge, Haoyang Li, Nicole Hu, Yongqi Zhang, Zhiyuan Wen, Jason Chen Zhang, Qing Li, Lei Chen.
- GraphAgent: An Effective Knowledge-Guided GNN Model Selection System
  VLDB 2026
  Mingtao Zhang, Haoyang Li, **Yuming Xu**, Nicole Hu, Peng Cheng, Jason Chen Zhang, Qing Li, Lei Chen.
- Fast and Faithful: A Lightweight Spatio-Temporal GNN for Semi-Supervised Air Quality Forecasting with Inductive Capability
  IEEE BigData 2025
  Yuming Xu, Zhanchao Xu, Yaowen Liu, Xuejia Chen, Qi Chen, Mingtao Zhang, Zhuohan Ge, Haoyang Li, Jason Chen Zhang.
- A Survey of Large Language Models in Mental Health Disorder Detection on Social Media
  ICDE 2025 Workshop
  Zhuohan Ge, Nicole Hu, Darian Li, Yubo Wang, Shihao Qi, **Yuming Xu**, Han Shi, Jason Chen Zhang.
- STRIDE: Lightweight and Effective Link Prediction over Spatio-Temporal Relational Graphs
  VLDB Journal Submission
  Yuming Xu, Haoyang Li, Yiming Li, Alexander Zhou, Yang Yang, Nicole Hu, Hanmo Liu, Darian Li, Jason Chen Zhang, Lei Chen, Qing Li.
- OpenGLT: A Comprehensive Benchmark of Graph Neural Networks for Graph-Level Tasks
  VLDB Submission
  Haoyang Li, **Yuming Xu**, Alexander Zhou, Yongqi Zhang, Lei Chen, Qing Li.
- Can Graph Foundation Models Replace Graph Neural Networks? An Experimental Study with Theoretical Analysis
  VLDB Journal Submission
  Mingtao Zhang*, **Yuming Xu***, Haoyang Li, Xinyi Zhu, Yiming Li, Yubo Wang, Alexander Zhou, Wei Dong, Jason Chen Zhang, Lei Chen, Qing Li (* Co-first author)

Research Interests

Graph Neural Networks, Graph Foundation Models, RAG Security, Agent Orchestration, AI for Science

Professional Service

Reviewer: *The Web Conference (WWW)*; *IEEE Transactions on Computational Social Systems*; *IEEE Transactions on Emerging Topics in Computational Intelligence*